

CSCE 676 DATA MINING · MODULE 05 · LECTURE NOTES

Dimensionality Reduction

Fewer dimensions, and what each method keeps.

MODULE OUTCOMES

- Explain the curse of dimensionality concretely.
- Explain PCA and SVD and what they preserve.
- Explain random projection and its guarantee.
- Explain the nonlinear methods and their dangers.
- Choose a method by what must be preserved.

1 The curse of dimensionality

- **Distances concentrate.** The ratio between the distance to the nearest and the farthest neighbour approaches 1 as the dimension grows — so 'nearest' stops being a meaningful notion, which breaks nearest-neighbour search, clustering (Module 04), and anomaly detection (Module 08) simultaneously.
- **Volume moves to the corners.** Almost all the volume of a high-dimensional ball lies near its surface, and the ball occupies a vanishing fraction of its bounding cube — so two-dimensional and three-dimensional intuitions mislead systematically rather than occasionally.
- **Sampling density collapses.** Covering a space to a fixed resolution requires exponentially many points in the dimension, which means any real dataset is extremely sparse in its own space.
- **And everything becomes orthogonal.** Two random vectors in high dimension are nearly perpendicular with high probability — which is both a problem and the basis of random projection (§3), and that dual role is worth noticing.
- **So reduction is not only about computational cost** — it is frequently about making the problem meaningful at all. Distance concentration is the consequence with the widest reach, because so many of this course's methods are distance-based.

BUT REAL HIGH-DIMENSIONAL DATA USUALLY HAS LOW INTRINSIC DIMENSION

A thousand-pixel image of a single object rotating varies along exactly one parameter — so the data lies on a one-dimensional curve embedded in a thousand-dimensional space, and 999 of those dimensions are unused.

And this is typical rather than contrived. Correlated features, physical constraints, and the generative process behind the data all mean that real data occupies a small part of its nominal space.

Which is exactly what makes reduction possible rather than merely lossy — you are discarding dimensions the data does not use, rather than compressing dimensions it does, and the distinction determines whether the reduction costs you anything.

So the question to ask of every method is: what structure does it preserve, and is that the structure you need? — which is §4's framing and this module's organising question, and it is a better way to choose than by popularity.

2 Linear methods

$$\mathbf{A} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^T \quad \mathbf{A}_k = \mathbf{U}_k \mathbf{\Sigma}_k \mathbf{V}_k^T$$

- Any matrix decomposes this way. $\mathbf{U}$ and $\mathbf{V}$ are orthogonal, and $\mathbf{\Sigma}$ is diagonal with the singular values in decreasing order — which is what makes truncation natural.
- Keeping the top k gives the best rank- k approximation in Frobenius norm — the Eckart–Young theorem, and this is a theorem rather than a heuristic, which is precisely what distinguishes these methods from the nonlinear ones in §4.
- And PCA is the SVD of the centred data — so they are the same computation, with PCA's covariance eigenvectors being the right singular vectors. Knowing that saves learning two things.
- Stating Eckart–Young explains why PCA is the default: it is not merely a reasonable choice but the provably optimal linear one for reconstruction error, which is a strong position for a method this cheap.
- Centre the data, and think carefully about scaling. PCA on unscaled features is dominated by whichever feature has the largest units — which is a measurement artefact rather than structure, and is the commonest mistake with the method.
- The explained variance ratio tells you how much you kept — and a scree plot is the honest way to choose k , with exactly the same elbow problem as Module 04 §4's: frequently there is no bend.
- Components are orthogonal and are not interpretable by default — reading meaning into principal component two is a common and unjustified move. PCA maximises variance, and variance has no reason at all to align with anything meaningful, so interpretation requires separate evidence.
- Truncated and randomised SVD scale to large matrices, computing only the top k components without forming the full decomposition — which is what makes this usable at this course's scale and is worth knowing exists.

- **And for sparse non-negative data, consider non-negative matrix factorisation**, whose components are additive rather than signed and are **frequently genuinely more interpretable** — at the cost of losing the optimality guarantee.

3 Random projection

PROJECTING ONTO RANDOM DIRECTIONS PRESERVES ALL PAIRWISE DISTANCES, WITH HIGH PROBABILITY

Any n points in any dimension whatsoever can be projected into $O(\log n / \epsilon^2)$ dimensions with every pairwise distance preserved to within a factor of $(1 \pm \epsilon)$ — the Johnson–Lindenstrauss lemma.

And the target dimension depends on n and ϵ only — not on the original dimension at all, which is the genuinely surprising part: a million-dimensional dataset and a thousand-dimensional one with the same number of points need the same target dimension.

The projection is simply a random matrix, so it costs nothing to construct, requires no pass over the data to fit, and is trivially parallel and streamable — all of which are properties PCA lacks, since PCA must be computed from the data.

So the trade against PCA is clear: PCA is optimal for your specific data and requires computing it; Johnson–Lindenstrauss is data-oblivious and gives a distance guarantee instead — and for distance-based methods, the distance guarantee is precisely what you wanted rather than a consolation. The proof is a concentration argument (CSCE 658 Module 03).

4 Nonlinear methods

Method	What it preserves	The danger
t-SNE	Local neighbourhoods — which points are near which.	Cluster sizes and inter-cluster distances are meaningless — see the note.
UMAP	Local structure, and somewhat more global structure than t-SNE.	The same caution, slightly less severely — and the hyperparameters change the picture substantially.
Isomap and MDS	Geodesic distances along a neighbour graph, or the metric directly.	$O(n^2)$, and very sensitive to the neighbour graph's construction — a single bad edge short-circuits the manifold.
Autoencoders	Whatever the loss function rewards.	No guarantee of any kind; and it needs training, tuning, and its own evaluation (CSCE 636).

The t-SNE caution is the important one, and it is worth attacking directly because the misreading is so common: in a t-SNE plot, the distance between two clusters and the relative sizes of the clusters carry no information at all — the method optimises local neighbourhood preservation and explicitly does not preserve either — and both are routinely interpreted in published figures, usually as evidence of how distinct or how large a group is. The perplexity parameter also changes the apparent number of clusters, which means a t-SNE figure without its parameters is uninterpretable.

CHOOSE BY WHAT MUST BE PRESERVED

Variance and a reconstruction guarantee → **PCA or truncated SVD** — which is also the right choice when you need to project new points cheaply, since the transformation is a fixed matrix.

Pairwise distances, with a stated bound, at scale → **random projection (§3)** — **especially as a preprocessing step before a distance-based method** such as LSH (Module 02) or clustering (Module 04).

Visual exploration of local structure → **t-SNE or UMAP, with the caveats stated on the figure itself** — **which almost nobody does**, and which would prevent most of the misreading.

And a learned nonlinear representation for a downstream task → **an autoencoder, evaluated by the downstream task rather than by its reconstruction loss** — since a low reconstruction error says nothing about whether the representation is useful for anything.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Tan et al., the dimensionality reduction appendix	§1 and §2 , with the curse of dimensionality developed carefully.
Mining of Massive Datasets, chapter 11 (free)	§2 and §3 — SVD, CUR, and the scale considerations, free in full.
Wattenberg, Viégas & Johnson — How to Use t-SNE Effectively (free)	§4's caution, demonstrated interactively — the single best corrective to the common misreading.
Achlioptas — Database-friendly random projections (free)	§3 with a projection matrix you can compute cheaply — the practical version of the lemma.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Measure the nearest-to-farthest distance ratio** as dimension grows from 2 to 1000.
- 2 **Compute the fraction of a ball's volume** within 10% of its surface, by dimension.
- 3 **Measure the angle between random vectors** at increasing dimension.
- 4 **Estimate the intrinsic dimension** of a real dataset you have.
- 5 **Run PCA with and without feature scaling** and compare the first component.
- 6 **Plot a scree plot** and say whether there is an elbow.

- 7 **Implement random projection** and measure the distance distortion against the bound.
- 8 **Compare PCA and random projection** as preprocessing for nearest-neighbour search.
- 9 **Run t-SNE at three perplexities** and compare the apparent cluster count.
- 10 **Write the caveat** you would put on a t-SNE figure.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Give four consequences of high dimension, and the one with widest reach.
- 2 Why does real data usually have low intrinsic dimension?
- 3 What makes reduction possible rather than merely lossy?
- 4 State the SVD and what truncation gives you.
- 5 What is the relationship between PCA and SVD?
- 6 Why are PCA components not interpretable?
- 7 State the Johnson–Lindenstrauss result and what is surprising about it.
- 8 Give the trade between PCA and random projection.
- 9 Name four nonlinear methods and what each preserves.
- 10 What carries no information in a t-SNE plot?